Chapter 3 · Semester 1

DYNAMICS

Notes by Cikgu Suhaili

24 worked examples 3 simulations Exercises with answers

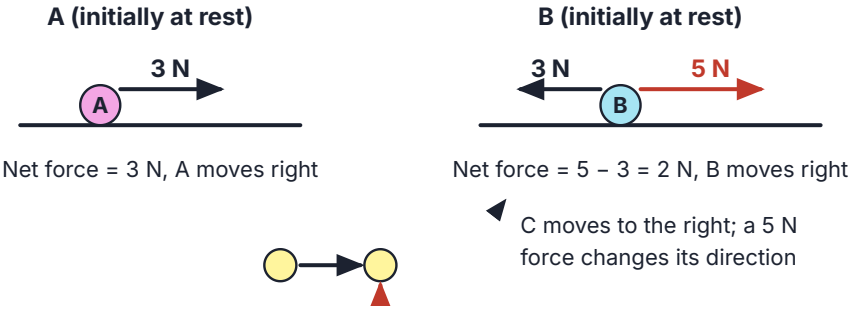
3.1 Newton's laws of motion

Newton's first law

Newton's first law

Every body **continues in its state of rest or uniform motion** in a straight line unless a **net force** acts on it. In equation form, $\Sigma F = 0$.

- A stationary particle starts to move because a net external force acts on it.
- A moving particle stops, or changes direction, because of a net external force.
- The first law tells us what a force does: it changes the state of motion.



Inertia

is **the tendency of a body to maintain its state of rest or of uniform motion in a straight line**. Newton's first law is also called the law of inertia.

- Inertia is a body's reluctance to start moving ("lazy") and its reluctance to stop once it is moving ("hardworking"), so a force is needed to move a body or to stop it.
- Inertia depends on **mass**: **mass is a measure of the inertia of a body**. A heavy body is harder to move, and harder to stop, than a light one.
- When a car stops suddenly the passenger continues to move forward because of his inertia, so a safety belt is needed during emergency braking.
- Classic inertia experiments: pulling a sheet from under a pile of books, and pushing two hanging tins, one empty and one full of sand.

Newton's second law

Newton's second law

The **rate of change of momentum** is **proportional to the resultant force** (net force).

$$\frac{\mathrm{d}p}{\mathrm{d}t} \propto F \Rightarrow \frac{mv - mu}{t} \propto F \Rightarrow ma \propto F \Rightarrow F = kma$$

- If m = 1 kg, a = 1 m s^{-2} and k = 1, then F = 1 N.
- The **direction of the acceleration** is the same as the direction of the applied net force.
- **1 newton** is the force that produces an acceleration of 1 m s^{-2} when acting on a 1 kg mass: **1 N = 1 kg m s^{-2}**.

$$F = \frac{\mathrm{d}(mv)}{\mathrm{d}t} = v\frac{\mathrm{d}m}{\mathrm{d}t} + m\frac{\mathrm{d}v}{\mathrm{d}t}$$

If the mass does not change, $\frac{\mathrm{d}m}{\mathrm{d}t} = 0$ and the acceleration is $a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{v - u}{t}$, so

$$F = ma \quad \text{or} \quad F = \frac{m(v - u)}{t}$$

Example 3.1.1

The velocity of a particle of mass **200 g** moving in a straight line increases from **10 to 50 cm s^{-1} in 2 s**. Find the **force** acting on the particle.

Solution

m = 200 × 10^{-3} kg, v = 50 × 10^{-2} m s^{-1}, u = 10 × 10^{-2} m s^{-1}

$$F = \frac{m(v - u)}{t} = \frac{200 \times 10^{-3}[(50 - 10) \times 10^{-2}]}{2} = \mathbf{4.0 \times 10^{-2}\ N}$$

Example 3.1.2

A sprinter accelerates constantly from **rest** and covers **50 m** in **6 s**. Her mass is **54 kg**. Ignoring all resistance, what **horizontal force** is she producing?

Solution

m = 54 kg, u = 0, s = 50 m, t = 6 s

$s = ut + \frac{1}{2}at^2 \Rightarrow a = \frac{s - ut}{\frac{1}{2}t^2} = \frac{50 - 0}{\frac{1}{2}(6)^2} = 2.8 \text{ m s}^{-2}$

$F = ma = (54)(2.8) =$ **151 N**

Example 3.1.3

At time t seconds, the acceleration of a body of mass 5 kg is $(10 + 70t - 20t^2)$ m s^{-2}. Find the **force** on the body at t **= 0**, and the **maximum force** on the body.

Solution

At t = 0: $a = 10 + 0 - 0 = 10$ m s^{-2}, so $F = ma = (5)(10) =$ **50 N**

Maximum force: $F_{max} = ma_{max}$. For a_{max}, $\frac{da}{dt} = 0$: $70 - 2(20)t = 0 \Rightarrow t = 1.75$ s

$a = 10 + 70(1.75) - 20(1.75)^2 = 71.25$ m s^{-2}

$F_{max} = 5(71.25) =$ **356.3 N**

Using $F = m\,dv/dt + v\,dm/dt$ when the velocity is constant

When the **velocity is constant**, $dv/dt = 0$ and the force is needed only to change the mass that is moving:

$$F = v\frac{dm}{dt}$$

dm/dt = rate at which mass is added to (or removed from) the moving stream

Additional example (syllabus 3.1(d)) · conveyor belt

Sand falls vertically onto a conveyor belt at a rate of 2.0 kg s^{-1}. The belt moves horizontally at a constant speed of 3.0 m s^{-1}. Find the extra horizontal force needed to keep the belt moving at this speed.

Solution

The sand lands with no horizontal velocity and must be brought up to 3.0 m s^{-1}. The belt's speed is constant, so $dv/dt = 0$.

$$F = m\frac{dv}{dt} + v\frac{dm}{dt} = 0 + (3.0)(2.0) = \mathbf{6.0\ N}$$

Additional example (syllabus 3.1(d)) · water jet

Water leaves a hose at 20 m s^{-1} at a rate of 0.50 kg s^{-1} and hits a wall horizontally without bouncing back. Find the force exerted by the water on the wall.

Solution

Each second, 0.50 kg of water changes its velocity from 20 m s^{-1} to 0. The jet speed is constant, so the force is $F = v\frac{dm}{dt}$.

$F = (20)(0.50) = 10$ N. The wall exerts 10 N on the water, so by Newton's third law **the water exerts 10 N on the wall**, in the direction of the jet.

Exercises: F = ma

1. A lorry of mass 5000 kg can accelerate at 1 m s^{-2} when it is unloaded. What will its acceleration be when it carries a load of 3000 kg?

   Answer

   0.625 m s^{-2}

   FULL SOLUTION

   The driving force F is the same in both cases.

   Unloaded: F = ma = 5000 × 1 = 5000 N

   Loaded: total mass = 5000 + 3000 = 8000 kg

   a = F/m = 5000 / 8000

= **0.625 m s^{-2}**

2. A car of mass 1000 kg experiences a net horizontal force of 4000 N, which accelerates it from rest to 10 m s^{-1}. How far does it travel during the acceleration?

Answer

12.5 m

FULL SOLUTION

a = F/m = 4000 / 1000 = 4.0 m s^{-2}

u = 0, v = 10 m s^{-1}: $v^2 = u^2 + 2as$

s = $(10^2 - 0) / (2 \times 4.0)$

= **12.5 m**

3. A car of mass 800 kg experiences a force of 1200 N. How long does it take to reach 30 m s^{-1} from rest?

Answer

20 s

FULL SOLUTION

a = F/m = 1200 / 800 = 1.5 m s^{-2}

v = u + at ⇒ 30 = 0 + 1.5t

t = **20 s**

4. At time t seconds, the force on a body of mass 10 kg is $(4t - t^2)$ N. Find the acceleration at t = 1 s and the maximum acceleration.

Answer

0.3 m s^{-2}; 0.4 m s^{-2} (at t = 2 s)

FULL SOLUTION

a = F/m = $(4t - t^2) / 10$

At t = 1 s: a = $(4 - 1)/10$ = **0.3 m s^{-2}**

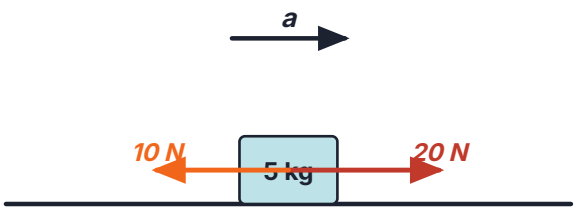
Maximum a: da/dt = $(4 - 2t)/10 = 0$ ⇒ t = 2 s

($d^2a/dt^2 = -0.2 < 0$, so this is a maximum)

$a_{max} = (4(2) - 2^2)/10$ = **0.4 m s^{-2}** (at t = 2 s)

Example 3.1.4

Two forces act on an object P of mass 5 kg: 20 N to the right and 10 N to the left. Find the **acceleration**.



Solution

Net force, F = 20 − 10 = 10 N to the right

$F = ma$: 10 = 5a ⇒ a = **2 m s^{-2}** (to the right)

Exercises: net force and acceleration

1. Calculate the acceleration of a 10.0 kg box that moves horizontally when acted on by: (a) 3 N and 5 N, both to the right; (b) 3 N at 45° above the horizontal and 5 N, both towards the right; (c) 3 N vertically upward and 5 N to the right; (d) 3 N at 45° above the horizontal to the right and 5 N to the left.

   Answer

   (a) 0.8 m s^{-2} to the right (b) 0.71 m s^{-2} to the right (c) 0.5 m s^{-2} to the right (d) 0.29 m s^{-2} to the left

   Only horizontal components accelerate a box that moves horizontally: (b) 3 cos 45° + 5 = 7.12 N; (c) the vertical 3 N does not count, so 5 N; (d) 5 − 3 cos 45° = 2.88 N.

   FULL SOLUTION

   Only horizontal components accelerate a box that moves horizontally. m = 10.0 kg.

   (a) F = 3 + 5 = 8 N ⇒ a = 8/10.0 = **0.8 m s^{-2} to the right**

   (b) F = 3 cos 45° + 5 = 2.12 + 5 = 7.12 N ⇒ a = 7.12/10.0 = **0.71 m s^{-2} to the right**

   (c) The vertical 3 N has no horizontal component: F = 5 N ⇒ a = 5/10.0 = **0.5 m s^{-2} to the right**

   (d) F = 5 − 3 cos 45° = 5 − 2.12 = 2.88 N to the left ⇒ a = 2.88/10.0 = **0.29 m s^{-2} to the left**

2. Find the acceleration of (a) a 6 kg body pulled by 12 N along a smooth floor; (b) a body of weight 49 N pulled up by two vertical forces of 12 N and 30 N.

Answer

(a) 2 m s^{-2} to the right (b) 1.4 m s^{-2} downward

FULL SOLUTION

(a) Smooth floor, so the net force is 12 N: a = F/m = 12/6

= **2 m s^{-2}** in the direction of the pull

(b) m = W/g = 49/9.81 = 5.0 kg

Net upward force = 12 + 30 − 49 = −7 N (i.e. 7 N downward)

a = 7 / 5.0 = **1.4 m s^{-2} downward**

3. Find the unknown forces: (a) force A gives a 6 kg body an acceleration of 3 m s^{-2} on a smooth floor; (b) force C lifts a 20 kg body (weight 196 N) with an upward acceleration of 1 m s^{-2}.

Answer

(a) A = 18 N (b) C = 216 N

FULL SOLUTION

(a) A = ma = 6 × 3

= **18 N**

(b) Taking upward as positive: C − mg = ma

C = 196 + 20 × 1

= **216 N**

Newton's third law

Newton's third law

For **every action (force)** there is a **reaction (opposing force)** of **equal magnitude but opposite direction**.

- If object A exerts a force on object B ($\mathbf{F}_{AB}$), then B exerts a force of the same size on A ($\mathbf{F}_{BA}$), in the opposite direction.
- The action and reaction act on **different bodies**.

$$\mathbf{F}_{AB} = -\mathbf{F}_{BA} \quad \text{(action)} \quad \text{(reaction)}$$

- A hand pushes on a wall (action); the wall pushes on the hand (reaction).
- A hand presses on a desk (action); the desk pushes up on the hand (reaction).
- We can walk forward because, when one foot pushes backward on the ground, the ground pushes forward on the foot.

3.2 Conservation of linear momentum and impulse

Linear momentum, p

is the product of the **mass** of a particle, m, and its **velocity**, v: $p = mv$.

- A **vector**, in the same direction as the velocity.
- Unit: kg m s^{-1} or N s.

Principle of conservation of linear momentum

When the **net external force** on a system is **zero**, the total **momentum** of the system is constant. Or: if there are **no external forces** acting on a system, the total momentum before a collision **equals** the total momentum after it.

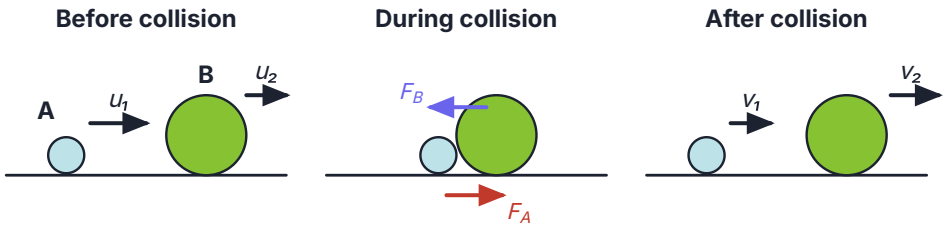
Newton's second law: $F = \frac{\Delta p}{\Delta t}$. If $\Sigma F = 0$ then $\Delta p = 0$, so p = constant

$$\Sigma p_i = \Sigma p_f$$

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

For a collision between two bodies

Newton's third law proves the principle



During the collision: $F_A = m_1\frac{v_1 - u_1}{t}$ and $F_B = m_2\frac{v_2 - u_2}{t}$

By Newton's third law, $F_A = -F_B$ (the minus sign: F_B is opposite to F_A), so $m_1\frac{v_1 - u_1}{t} = -m_2\frac{v_2 - u_2}{t}$

Rearranging: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$. With no external force, Newton's third law proves the principle of conservation of linear momentum.

Elastic and inelastic collisions

Elastic collision
is one in which the **total kinetic energy is conserved**.

- Total momentum is conserved.
- Total kinetic energy is conserved.

Inelastic collision
is one in which the **total kinetic energy is not conserved**.

- Total momentum is conserved.
- Total kinetic energy is not conserved (some is changed to heat, sound and deformation).
- In a **perfectly inelastic** collision the bodies stick together after the collision.

Elastic collision in one dimension (head-on)

Momentum: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

Kinetic energy: $\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$

If m_2 is at rest ($u_2 = 0$): from momentum, $m_1(u_1 - v_1) = m_2v_2$ **(1)**

From kinetic energy, $m_1(u_1^2 - v_1^2) = m_2v_2^2$ **(2)**

(2) ÷ (1): $u_1 + v_1 = v_2$ **(3)**

Solving (3) and (1):

$$v_1 = \frac{m_1 - m_2}{m_1 + m_2}u_1 \qquad v_2 = \frac{2m_1}{m_1 + m_2}u_1$$

Elastic collision, m_2 initially at rest

Masses	After the collision
$m_1 = m_2$	$v_1 = 0$, $v_2 = u_1$ (they exchange velocities)
$m_1 > m_2$	$v_1 < u_1$; $v_2 > u_1$: m_1 slows down
$m_1 < m_2$	$v_1 < 0$; $v_2 < u_1$: m_1 recoils
$m_1 \gg m_2$	$v_1 \approx u_1$; $v_2 \approx 2u_1$
$m_1 \ll m_2$	$v_1 \approx -u_1$; $v_2 \approx 0$

Example 3.2.1

A 3000 kg truck moving at 10 m s^{-1} hits a 1000 kg parked car. The impact sets the car in motion at 15 m s^{-1}. Determine the **velocity** of the truck after the collision.

Solution

$(\Sigma p_{\text{before}})_x = (\Sigma p_{\text{after}})_x$: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$(3000)(10) + (1000)(0) = (3000)v_1 + (1000)(15) \Rightarrow 30\,000 - 15\,000 = 3000v_1$

$v_1 = 5.0$ **m s^{-1}** (to the right)

Check with the elastic-collision formula: $v_1 = \dfrac{3000 - 1000}{3000 + 1000}(10) = 5.0$ m s^{-1}

Perfectly inelastic collision in one dimension

After the collision the two objects stick together. By conservation of momentum:

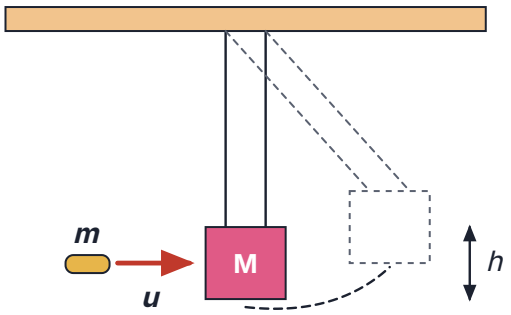
$$m_1u_1 + m_2u_2 = (m_1 + m_2)v \;\Rightarrow\; v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$$

But kinetic energy is **not conserved**: $\tfrac{1}{2}m_1u_1^2 + \tfrac{1}{2}m_2u_2^2 \neq \tfrac{1}{2}(m_1 + m_2)v^2$. If m_2 is at rest:

$$\text{K.E.}_i = \tfrac{1}{2}m_1u_1^2, \quad \text{K.E.}_f = \tfrac{1}{2}(m_1 + m_2)v^2, \quad \frac{\text{K.E.}_f}{\text{K.E.}_i} = \frac{m_1}{m_1 + m_2} < 1$$

Ballistic pendulum

Ballistic pendulum
is a **device which measures the speed of a bullet**.



- M = mass of the block; m = mass of the bullet
- u = velocity of the bullet before the collision
- v = velocity of the block and bullet just after the collision
- h = maximum height reached by the block containing the bullet

The block is at rest. By conservation of momentum (perfectly inelastic): $\boldsymbol{mu = (M + m)v}$ (i)

After the collision, by conservation of energy (kinetic energy → gravitational potential energy): $\frac{1}{2}(M + m)v^2 = (M + m)gh \Rightarrow \boldsymbol{v = \sqrt{2gh}}$ (ii)

Substitute (ii) into (i) to get the speed of the bullet:

$$u = \frac{m + M}{m}\sqrt{2gh}$$

Momentum is conserved during the collision (it is too quick for the block to rise), but kinetic energy is not. Energy is conserved only afterwards, while the block swings up.

Example 3.2.7

In a ballistic pendulum experiment, h = 5.00 cm, m = 5.00 g and M = 1.00 kg. Find (a) the initial speed of the bullet, u, (b) the loss in kinetic energy due to the collision.

Solution

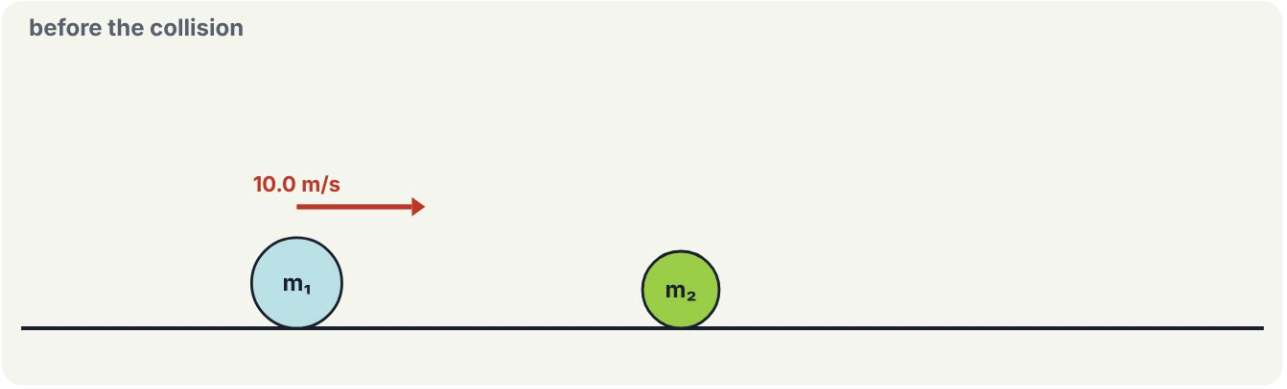
(a) $u = \dfrac{m + M}{m}\sqrt{2gh} = \dfrac{(5 \times 10^{-3}) + 1}{5 \times 10^{-3}}\sqrt{2(9.81)(5 \times 10^{-2})} = \mathbf{199\ m\ s^{-1}}$

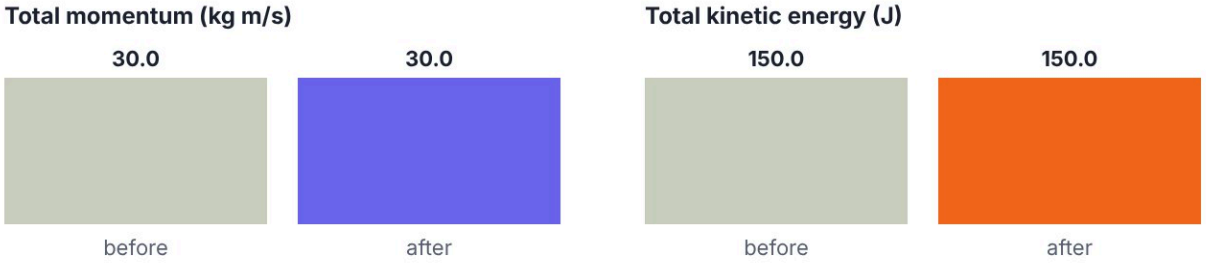
(b) $\Delta K = K_i - K_f = \tfrac{1}{2}mu^2 - \tfrac{1}{2}(m + M)v^2 = \tfrac{1}{2}mu^2 - \tfrac{1}{2}(m + M)(2gh)$

$= \tfrac{1}{2}(5 \times 10^{-3})(199)^2 - \tfrac{1}{2}(5 \times 10^{-3} + 1.00)(2)(9.81)(5 \times 10^{-2}) = 99.0 - 0.5 = \mathbf{98.5\ J}$

Simulation **Collisions: elastic, inelastic and perfectly inelastic**

Set the two masses and their velocities, choose the type of collision, then press Play. The bars show that total momentum is always the same before and after, while kinetic energy is conserved only in an elastic collision.





v_1 after	v_2 after	momentum before → after	KE lost
5.00 m/s	**15.00 m/s**	**30.0 → 30.0**	**0.0 J (0%)**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 3 to use the simulation.

Example 3.2.3

A 1800 kg truck stopped at a traffic light is struck from the rear by a 900 kg car and the two become entangled. If the car was moving at 20 m s^{-1} before the collision, what is the speed of the entangled mass after the collision?

Solution

m_1 = 900 kg, u_1 = 20 m s^{-1}; m_2 = 1800 kg, u_2 = 0

$\Sigma p_i = \Sigma p_f$: $m_1u_1 = (m_1 + m_2)v$

(900)(20) = (1800 + 900)v ⇒ **v = 6.67 m s^{-1}**

Exercises: perfectly inelastic collisions

1. **Example 3.2.4** A truck of mass 3000 kg travelling at 3 m s^{-1} runs into a stationary truck of mass 5000 kg. The trucks couple together. Find their common speed.

   Answer

   1.13 m s^{-1}

   FULL SOLUTION

   Momentum is conserved: $m_1u_1 + m_2u_2 = (m_1 + m_2)v$

   3000(3) + 5000(0) = (3000 + 5000)v

   v = 9000 / 8000 = 1.125 = **1.13 m s^{-1}**

2. **Example 3.2.5** A 16 g mass moves in the +x-direction at 30 cm s^{-1} while a 4.0 g mass moves in the −x-direction at 50 cm s^{-1}. They collide head on and stick together. Find their velocity after the collision.

   Answer

   0.14 m s^{-1} in the +x-direction

   FULL SOLUTION

   Take +x as positive: m_1 = 0.016 kg, u_1 = +0.30 m s^{-1}; m_2 = 0.0040 kg, u_2 = −0.50 m s^{-1}

   $m_1u_1 + m_2u_2 = (m_1 + m_2)v$

   0.016(0.30) + 0.0040(−0.50) = 0.020v

   0.0048 − 0.0020 = 0.020v ⇒ v = 0.14 m s^{-1}

   0.14 m s^{-1} in the +x-direction

3. **Example 3.2.6** A railway engine of mass 20 000 kg moving at 3 m s^{-1} collides with and couples to a stationary train of wagons of total mass 10 000 kg. Calculate (a) the speed just after impact, (b) the loss of kinetic energy.

Answer

(a) 2 m s^{-1} (b) 30 000 J

FULL SOLUTION

(a) 20 000(3) + 10 000(0) = (20 000 + 10 000)v

v = 60 000 / 30 000 = **2 m s^{-1}**

(b) K before = ½(20 000)$(3)^2$ = 90 000 J

K after = ½(30 000)$(2)^2$ = 60 000 J

Loss of kinetic energy = 90 000 − 60 000 = **30 000 J**

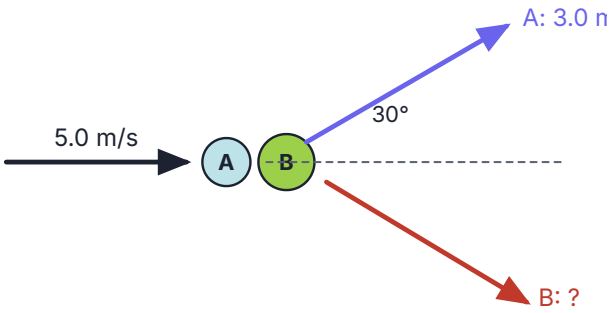
Collisions in two dimensions

Momentum is a vector, so in two dimensions it is conserved **separately in the x-direction and in the y-direction**:

$$\Sigma p_x \text{ before} = \Sigma p_x \text{ after} \quad \Sigma p_y \text{ before} = \Sigma p_y \text{ after}$$

Additional example (syllabus 3.3(c)) · collision in two dimensions

A ball A of mass 2.0 kg moving east at 5.0 m s^{-1} collides with a stationary ball B of mass 3.0 kg. After the collision A moves at 3.0 m s^{-1} at 30° north of east. Find (a) the velocity of B after the collision, (b) whether the collision is elastic.



Solution

Take east as +x and north as +y. Before: p_x = (2.0)(5.0) = 10 kg m s^{-1}, p_y = 0

x: 10 = (2.0)(3.0 cos 30°) + 3.0v_{Bx} ⇒ $v_{Bx} = \dfrac{10 - 5.20}{3.0}$ = 1.60 m s^{-1}

y: 0 = (2.0)(3.0 sin 30°) + 3.0v_{By} ⇒ v_{By} = −1.00 m s^{-1}

(a) $v_B = \sqrt{1.60^2 + 1.00^2}$ = **1.89 m s^{-1}** at θ = $\tan^{-1}\dfrac{1.00}{1.60}$ = **32° south of east**

(b) K.E. before = ½(2.0)(5.0)^2 = 25 J; K.E. after = ½(2.0)(3.0)^2 + ½(3.0)(1.89)^2 = 9.0 + 5.3 = 14.3 J. Kinetic energy is not conserved, so the collision is **inelastic**.

Impulse of a force

Impulse

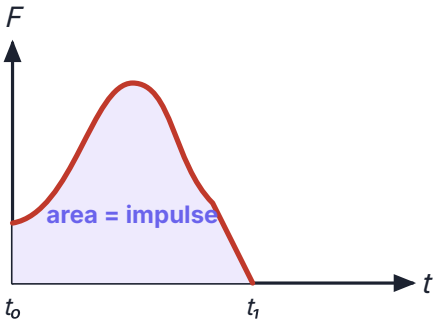
is the product of the **applied force**, F, and the **time interval** Δt for which it acts (for a constant force): $I = F\Delta t$.

From Newton's second law, $F = \frac{\Delta(mv)}{\Delta t} = \frac{\Delta p}{\Delta t}$

$$F\Delta t = \Delta p = m(v - u)$$

Impulse = change of momentum

- F is the impulsive force.
- A vector, in the direction of the force.
- Unit: kg m s^{-1} or N s.



Impulse = area under the graph of F against t = $\int_{t_0}^{t_1} F\,dt$

Applications

From $F\Delta t = m(v - u)$: $F = \frac{m(v - u)}{\Delta t}$

- To **increase** the impulsive force, the change of momentum must happen in a **very short time**: a karate chop, smashing a shuttlecock, kicking a ball, squash and golf.
- In a car accident, to **reduce** the impulsive force, the collision time is increased by using softer bodies (crumple zones, airbags).

Example 3.2.15

A car of mass 800 kg is travelling at 25 m s^{-1}. Find the **constant force** needed to stop it in 4 s.

Solution

u = 25 m s^{-1}, v = 0, t = 4 s, m = 800 kg

$F = \frac{m(v - u)}{t} = \frac{800(0 - 25)}{4}$ = −5000 N, so a force of **5000 N** opposite to the motion

Example 3.2.16

A ball of mass 0.2 kg is dropped from a height of 20 m. It rebounds to a height of 10 m. Find the **impulse** exerted by the ground on the ball.

Solution

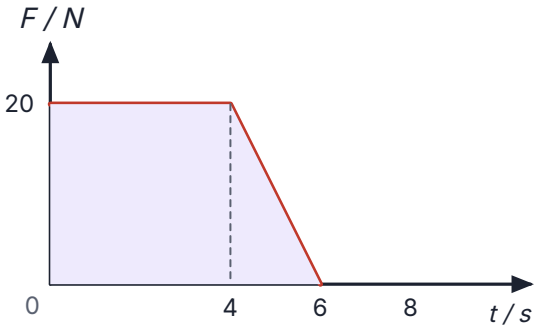
Speed just before impact: $u = \sqrt{2gh} = \sqrt{2(9.81)(20)} = 19.81$ m s^{-1} (downward)

Speed just after: $v = \sqrt{2(9.81)(10)} = 14.01$ m s^{-1} (upward)

Taking upward as positive: impulse = $m(v - u)$ = 0.2[14.01 – (−19.81)] = **6.76 N s** upward

Example 3.2.17

The force acting on an object is shown below. Find the impulse at (a) t = 6 s, (b) t = 8 s.



Solution

(a) Impulse = area = ½(4 + 6)(20) = **100 N s**

(b) No force acts after 6 s, so the total area is still 100 + 0 = **100 N s**

Seat belts, airbags and crumple zones

In a crash the passenger's momentum must fall from mv to zero: the **impulse** $F\Delta t = \Delta p$ is fixed by the speed. What safety features change is the **time** Δt over which it happens. A longer time means a **smaller force** on the body.

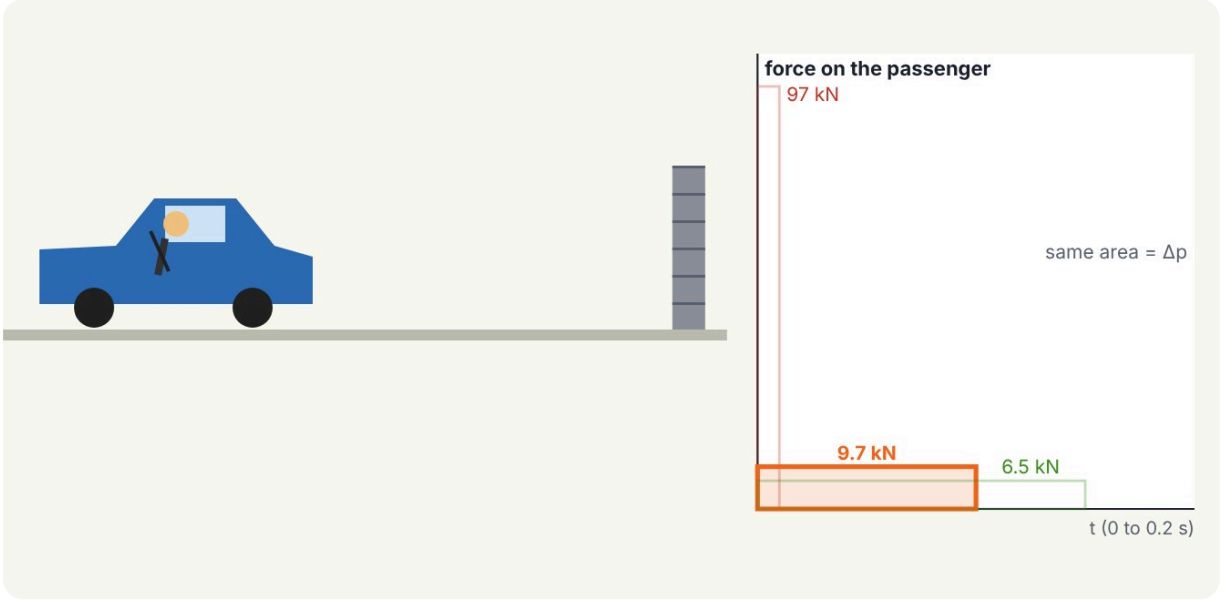
- **No seat belt:** the passenger keeps moving until they hit the dashboard or windscreen, and stop in about 0.01 s: an enormous force.
- **Seat belt:** it stretches slightly and stops the passenger together with the car over about 0.1 s.

- **Airbag:** a soft cushion that increases the stopping time further and spreads the force over a larger area.
- **Crumple zone:** the front of the car is designed to fold up, so the car itself takes longer to stop.

Simulation

Crash test: why seat belts and airbags save lives

Choose the speed and the safety system, then crash the car (it is shown slowed down). The change of momentum, and so the area under the force–time graph, is the same each time; a longer stopping time gives a much smaller force.



change of momentum Δp = mv	stopping time Δt	average force F = Δp/Δt	force ÷ body weight
972 N s	**100 ms**	**9.7 kN**	**14 ×**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 3 to use the simulation.

Example 3.2.18

A 70 kg passenger is in a car travelling at 50 km h^{-1} when it crashes. Find the average force on the passenger if they are stopped (a) by a seat belt in 0.10 s, (b) by hitting the dashboard in 0.010 s. Compare each with the passenger's weight.

Solution

v = 50 km h^{-1} = 13.9 m s^{-1}; $\Delta p = mv$ = (70)(13.9) = 972 N s

(a) $F = \dfrac{\Delta p}{\Delta t} = \dfrac{972}{0.10}$ = **9.7×10^3 N**, about 14 times the weight (687 N)

(b) $F = \dfrac{972}{0.010}$ = **9.7×10^4 N**, about 140 times the weight: enough to cause fatal injuries

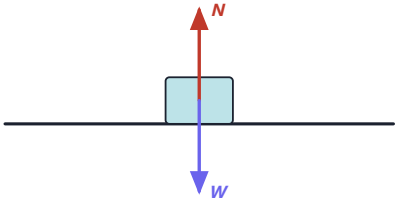
3.3 Reaction and frictional force

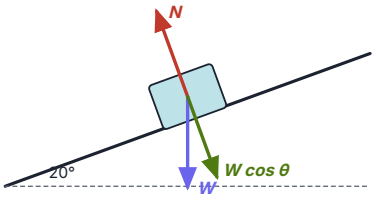
Normal (reaction) force, N or R

- Newton's third law example: the weight of an object presses on a table (action); the table pushes up on the object with a normal force (reaction).
- "Normal" means perpendicular.

Normal force

is the contact force exerted by the surface of one object on the surface of another, which prevents the objects from passing through each other. It acts **perpendicular to the surface**.



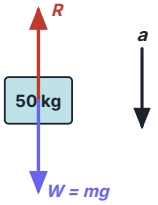


Free-body diagram

is a diagram of a body showing all the forces acting on it.

Example 3.3.1

A man of mass 50 kg stands on the floor of a lift which is descending with a downward acceleration of 0.8 m s^{-2}. Find the **reaction** of the floor of the lift on the man.



Solution

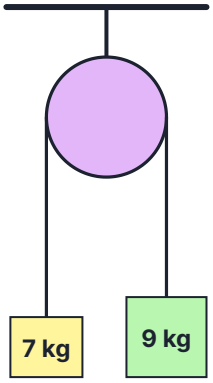
Free-body diagram: R upward, $W = mg$ downward, a downward

$\Sigma F_y = ma$ (downward positive): $ma = mg - R$

$R = mg - ma = 50(9.81 - 0.8) =$ **450.5 N**

Example 3.3.2

A cord passing over an easily turned pulley (massless and frictionless) has a 7.0 kg mass hanging from one end and a 9.0 kg mass from the other. Find the **acceleration** of the masses and the **tension** in the cord.



Solution

7.0 kg (accelerates up): $ma = T - mg \Rightarrow T = 7.0(a + 9.81)$ **(1)**

9.0 kg (accelerates down): $ma = mg - T \Rightarrow T = 9.0(9.81 - a)$ **(2)**

(1) = (2): a **= 1.23 m s^{-2}**, T **= 77.3 N**

Exercises: normal force and connected bodies

1. A force of 15.0 N is applied at 30° below the horizontal (pushing down on the block) on a 0.750 kg block resting on a frictionless surface. Calculate (a) the magnitude of the resulting acceleration, (b) the magnitude of the normal force.

Answer

(a) 17.3 m s^{-2} (b) 14.85 N

FULL SOLUTION

The force is 15.0 N at 30° below the horizontal; m = 0.750 kg; frictionless surface.

(a) Horizontal: F cos 30° = ma

a = 15.0 cos 30° / 0.750 = 12.99 / 0.750

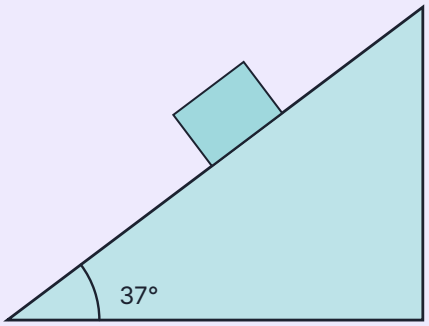
= **17.3 m s^{-2}**

(b) Vertical (no vertical acceleration): N = mg + F sin 30°

N = 0.750(9.81) + 15.0(0.5) = 7.36 + 7.50

= **14.9 N** (14.86 N)

2. A 10.0 kg mass is supported on a friction-free plane inclined at 37° and connected to a second mass M by an ideal string and pulley. M accelerates upward at 3.00 m s^{-2}. Calculate (a) the reaction force of the plane on the 10.0 kg mass, (b) the tension in the string, (c) the mass M.



Answer

(a) R = 78.3 N (b) T = 29.0 N (c) M = 2.27 kg

The slide answers 78.4 N, 28.8 N and 2.25 kg come from using g = 9.8 m s^{-2} with sin 37° ≈ 0.6 and cos 37° ≈ 0.8.

FULL SOLUTION

M accelerates upward, so the 10.0 kg mass slides down the plane with a = 3.00 m s^{-2}.

(a) Perpendicular to the plane (no acceleration): R = mg cos 37°

R = 10.0(9.81) cos 37°

= **78.3 N**

(b) For the 10.0 kg mass along the plane: mg sin 37° − T = ma

T = 10.0(9.81) sin 37° − 10.0(3.00) = 59.04 − 30.0

= **29.0 N**

(c) For M: T − Mg = Ma ⇒ M = T/(g + a)

M = 29.0 / (9.81 + 3.00)

= **2.27 kg**

Frictional force, f

Frictional force

is the contact force exerted **between the surfaces** of two objects, **directed parallel to the surface**, that acts to **prevent (oppose) the objects** from slipping on each other. It results from forces between atoms and molecules when surfaces are in contact.

- The frictional force is **proportional to the normal force**: $f \propto N$.
- Friction occurs when an object moves on a rough surface or through a fluid.
- Three types: (a) static friction, (b) kinetic friction, (c) rolling friction.

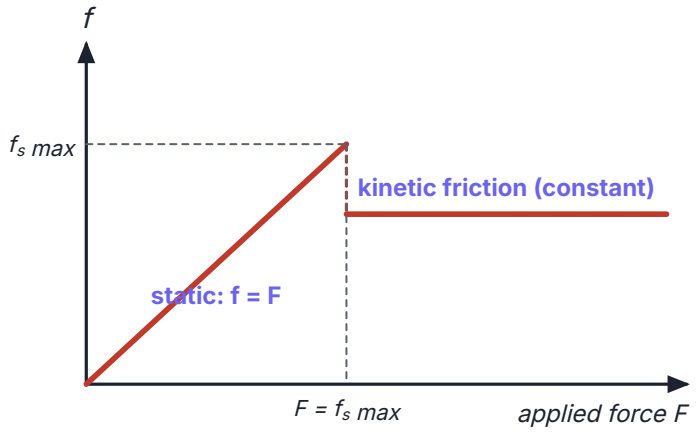
Static friction

- If the applied force F is increased gradually, at a certain value the book starts to move.
- The book just begins to move when F is slightly greater than the maximum static frictional force, f_s^{max}.
- The maximum frictional force produced by the rough surface is called the **limiting frictional force**.

Applied force	Static friction	Result
F = 3 N	f_s = 3 N	no movement
F = 5 N	f_s = 5 N	no movement
F = 7 N	f_s^{max} = 7 N	just about to move

$$f_s \le f_s^{max} \quad f_s^{max} = \mu_s N$$

μ_s = coefficient of static friction, N = normal force



Static friction grows with the applied force up to f_s^{max}; once the body slides, the kinetic friction is constant and smaller.

Simulation

Friction: push the block until it slides

Increase the applied force slowly. Static friction matches your push exactly until it reaches $\mu_s N$; then the block breaks away and the friction drops to $\mu_k N$. The graph of f against F builds up as you go.



normal force N = mg	limiting friction $\mu_s N$	kinetic friction $\mu_k N$	friction now	acceleration
19.6 N	**11.8 N**	**7.8 N**	**0.0 N**	**0**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 3 to use the simulation.

Coefficient of static friction, μ_s

is a **dimensionless constant, characteristic of the two surfaces**, which gives the relative strength of the force: $\mu_s = \dfrac{f_s^{max}}{N}$, where f_s^{max} is the limiting frictional force and N the normal force.

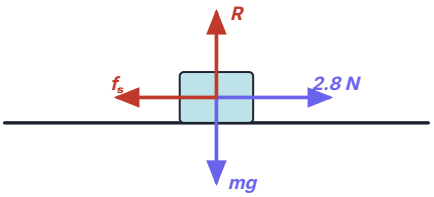
- μ_s depends on the **type** of material (wood on wood, rubber on concrete), the **condition** of the surfaces (polished, rough, lubricated) and other variables such as temperature.
- For smooth surfaces, $\mu_s = 0$.
- The maximum frictional force is **independent of the area of contact** if the normal reaction is the same.

Table 3.1 Coefficients of static and kinetic friction

Surfaces	Static, μ_s	Kinetic, μ_k
Rubber on dry concrete	1.0	0.7
Shoes on wood	0.9	0.7
Rubber on wet concrete	0.7	0.5
Steel on steel (dry)	0.6	0.3
Metal on wood	0.5	0.3
Wood on wood	0.4	0.2
Shoes on ice	0.1	0.05
Steel on steel (oiled)	0.05	0.03

Example 3.3.3

A body of mass 1.2 kg lies on a horizontal surface. A horizontal force of 2.8 N can just shift the body. Find the **coefficient of static friction**.



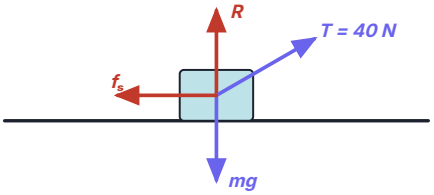
Solution

$\Sigma F_y = 0$: $R = mg$; $\Sigma F_x = 0$: $f_s = 2.8$ N

$f_s = \mu_s R = \mu_s mg \Rightarrow \mu_s = \dfrac{2.8}{(1.2)(9.81)} = \mathbf{0.24}$

Example 3.3.4

A body of mass 20 kg lies on a horizontal floor. A string attached to it at 30° to the horizontal has a tension of 40 N when the body begins to move. Find the coefficient of static friction.



Solution

$\Sigma F_y = 0$: $R + 40 \sin 30° = mg \Rightarrow R = mg - 40 \sin 30° = 176.2$ N

$\Sigma F_x = 0$: $40 \cos 30° = f_s = \mu_s R$

$$\mu_s = \frac{40 \cos 30°}{176.2} = \mathbf{0.197}$$

Kinetic friction

- Kinetic friction always **opposes the motion** of a moving object.
- It is less than the limiting static friction: the force needed to start motion is more than the force needed to keep it going, so $f_k < f_s^{max}$ and $\mu_k < \mu_s$.
- μ_k is independent of the velocity.

$$\mu_k = \frac{f_k}{N}, \quad f_k = \mu_k N$$

Example 3.3.5

A body is slid along a horizontal floor. It decelerates from 10 m s^{-1} to 5 m s^{-1} in 2 s. Find the **coefficient of friction** between the body and the floor.

Solution

$\Sigma F_y = 0$: $R = mg$; $\Sigma F_x = ma$: $ma = f_k = \mu_k mg$

$a = \dfrac{v - u}{t} = \dfrac{5 - 10}{2} = -2.5$ m s^{-2} (a deceleration of 2.5 m s^{-2})

$\mu_k = \dfrac{a}{g} = \dfrac{2.5}{9.81} = \mathbf{0.255}$

Example 3.3.6

A 70 kg box is slid along the floor by a 400 N force. The coefficient of kinetic friction is 0.50. Find the **acceleration** of the box.

Solution

$\Sigma F_y = 0$: $R = mg$; $\Sigma F_x = ma$: $ma = 400 - \mu_k mg$

$70a = 400 - (0.50)(70)(9.81) \Rightarrow \mathbf{a = 0.81\ m\ s^{-2}}$

Example 3.3.7

The 70 kg box is now pulled by a 400 N force at 30° to the horizontal. $\mu_k = 0.50$. Find the acceleration.

Solution

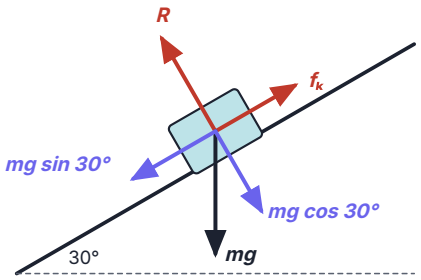
$\Sigma F_y = 0$: $R + 400 \sin 30° = mg \Rightarrow R = 486.7$ N

$\Sigma F_x = ma$: $ma = 400 \cos 30° - \mu_k R$

$70a = 346.4 - (0.50)(486.7) \Rightarrow \mathbf{a = 1.47\ m\ s^{-2}}$

Example 3.3.8

A 20 kg box sits on a 30° incline. μ_k between box and incline is 0.30. Find the acceleration of the box down the incline.



Solution

Perpendicular to the plane: $R = mg\cos 30° = 170$ N

Along the plane: $ma = mg\sin 30° - f_k = mg\sin 30° - \mu_k R$

$20a = (20)(9.81)(0.5) - (0.30)(170) \Rightarrow$ **$a = 2.36\ \text{m s}^{-2}$**

Exercises: friction on slopes and floors

1. A 10 kg block is placed on a plane inclined at 37°. μ = 0.4. When released, find (a) the normal force on the block, (b) its acceleration down the plane.

   Answer

   (a) 78.3 N (b) 2.77 m s^{-2}

   FULL SOLUTION
   (a) N = mg cos 37° = 10(9.81) cos 37°
   = **78.3 N**
   (b) Friction f = μN = 0.4 × 78.3 = 31.3 N (up the plane)
   Along the plane: mg sin 37° − f = ma
   10(9.81) sin 37° − 31.3 = 10a ⇒ 59.0 − 31.3 = 10a
   a = **2.77 m s^{-2}** down the plane

2. A 2 kg block is on a plane inclined at 30°; μ = 0.4. Find its acceleration (a) when pushed up the plane by a constant force of 20 N, (b) when pulled down the plane by the same force.

   Answer

   (a) 1.7 m s^{-2} (b) 11.51 m s^{-2}

FULL SOLUTION

mg sin 30° = 2(9.81)(0.5) = 9.81 N; N = mg cos 30° = 16.99 N; f = μN = 0.4 × 16.99 = 6.80 N

(a) Pushed up the plane: friction and the weight component both act down the plane.

20 − 9.81 − 6.80 = 2a ⇒ a = 3.39/2

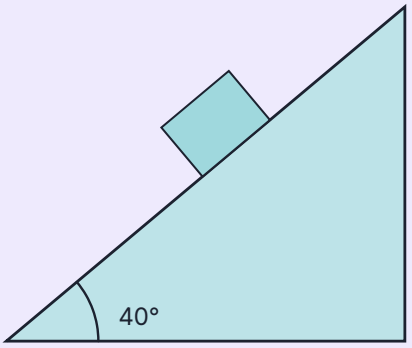
= **1.7 m s^{-2}** up the plane

(b) Pulled down the plane: friction acts up the plane, the weight component down the plane.

20 + 9.81 − 6.80 = 2a ⇒ a = 23.01/2

= **11.51 m s^{-2}** down the plane

3. A horizontal force of 500 N pushes a 25 kg box up a 40° incline with an acceleration of 0.75 m s^{-2}. Find the coefficient of kinetic friction. (g = 9.81 m s^{-2})



Answer

μ_k = 0.41

FULL SOLUTION

The 500 N force is horizontal, so it has components along and into the 40° incline.

Along the incline: 500 cos 40° = 383.0 N (up); weight component 25(9.81) sin 40° = 157.6 N (down)

Perpendicular: N = mg cos 40° + 500 sin 40° = 187.9 + 321.4 = 509.3 N

Up the incline: 500 cos 40° − mg sin 40° − μ_kN = ma

383.0 − 157.6 − μ_k(509.3) = 25(0.75) = 18.75

μ_k = (383.0 − 157.6 − 18.75) / 509.3

= **0.41**

4. A body on a rough 30° slope slides down with an acceleration of 4 m s^{-2}. Find the coefficient of friction.

Answer

0.106

FULL SOLUTION

Down the slope: mg sin 30° − μmg cos 30° = ma (the mass cancels)

g sin 30° − μg cos 30° = a

$9.81(0.5) - \mu(9.81)(0.866) = 4$

$\mu = (4.905 - 4) / 8.496$

$= \mathbf{0.106}$

5. A horizontal force of 140 N pulls a 60 kg box across a horizontal floor at constant speed. What is the coefficient of friction?

Answer

0.24

FULL SOLUTION

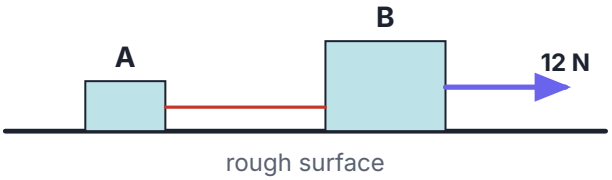
Constant speed ⇒ zero acceleration, so the pull equals the friction: $140 = \mu mg$

$\mu = 140 / (60 \times 9.81)$

$= \mathbf{0.24}$

Example 3.3.9

Blocks A (2 kg) and B (4 kg) are connected by a rope on a rough surface. The frictional forces on A and B are 2 N and 4 N. Find the acceleration and the tension in the rope if B is pulled by a force of 12 N.



rough surface

Solution

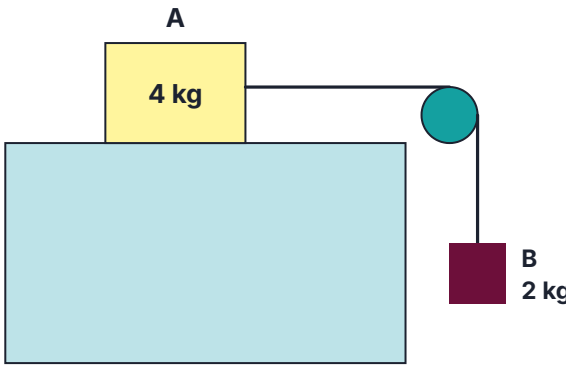
For A: $T - f_s = ma \Rightarrow T - 2 = 2a$ (1)

For B: $12 - T - f_s = ma \Rightarrow 8 - T = 4a$ (2)

(1) + (2): $6 = 6a \Rightarrow \mathbf{a = 1\ m\ s^{-2}}$; from (1): $\mathbf{T = 4\ N}$

Example 3.3.10

Object A (4 kg) on a rough table is connected to object B (2 kg) by a string over a pulley. The frictional force on A is 10 N. Find the acceleration, the tension in the string and μ_k.



Solution

For B: $m_B g - T = m_B a$; for A: $T - 10 = m_A a$

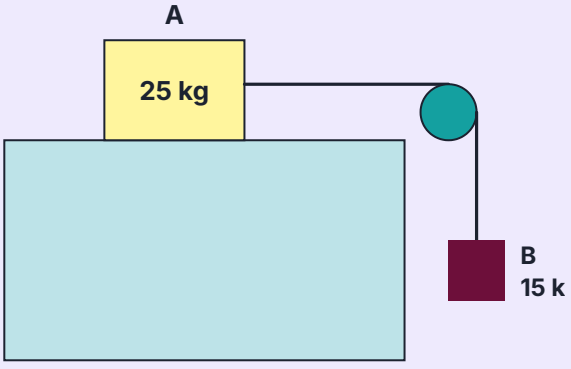
Adding: $2(9.81) - 10 = 6a \Rightarrow a = \mathbf{1.60\ m\ s^{-2}}$

$T = 2(9.81 - 1.60) = \mathbf{16.4\ N}$

$$\mu_k = \frac{f_k}{m_A g} = \frac{10}{4(9.81)} = \mathbf{0.25}$$

Exercises: connected bodies with friction

1. The coefficient of kinetic friction between block A (25 kg) and the table is 0.20. Block B (15 kg) hangs from a string over a pulley. How far will B drop in the first 3.0 s after the system is released?



Answer

11 m

FULL SOLUTION

Friction on A: $f = \mu m_A g = 0.20(25)(9.81) = 49.05$ N

For the whole system: $m_B g - f = (m_A + m_B)a$

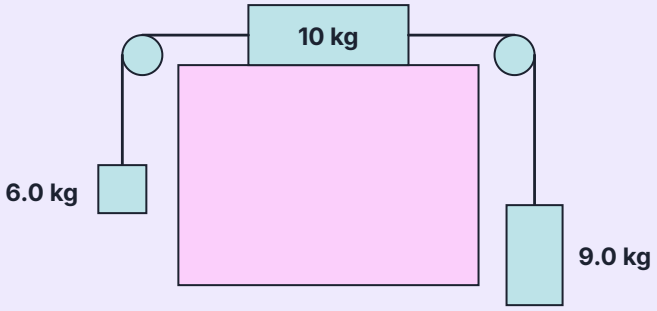
$15(9.81) - 49.05 = 40a \Rightarrow 147.15 - 49.05 = 40a$

$a = 2.45$ m s^{-2}

From rest: $s = \frac{1}{2}at^2 = \frac{1}{2}(2.45)(3.0)^2$

$= \mathbf{11\ m}$

2. Three objects are connected by strings over massless, friction-free pulleys: 6.0 kg hanging on the left, 10 kg on the table, 9.0 kg hanging on the right. μ_k between the middle object and the table is 0.2. (a) What is the acceleration? (b) Calculate the tension in each string.



Answer

(a) 0.39 m s^{-2} (b) 61 N and 85 N

FULL SOLUTION

The 9.0 kg mass is heavier, so it falls and the 6.0 kg mass rises.

Friction on the 10 kg object: $f = 0.2(10)(9.81) = 19.62$ N

(a) System: $9.0g - 6.0g - f = (6.0 + 10 + 9.0)a$

$88.29 - 58.86 - 19.62 = 25a \Rightarrow a = 9.81/25$

$= \mathbf{0.39\ m\ s^{-2}}$

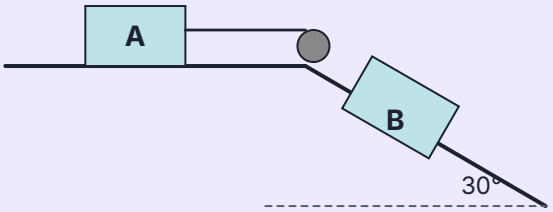
(b) Left string (6.0 kg rising): $T_1 - 6.0g = 6.0a \Rightarrow T_1 = 6.0(9.81 + 0.392)$

$T_1 = \mathbf{61\ N}$

Right string (9.0 kg falling): $9.0g - T_2 = 9.0a \Rightarrow T_2 = 9.0(9.81 - 0.392)$

$T_2 = \mathbf{85\ N}$

3. Two boxes of identical mass 40 kg are tied together: A on a horizontal surface, B on a 30° slope. Both experience sliding friction with $\mu_k = 0.15$. Find the acceleration and the tension in the tie cord.



Answer

1.1 m s^{-2}; 0.10 kN

FULL SOLUTION

B on the slope pulls A along the horizontal surface; $m_A = m_B = 40$ kg, $\mu_k = 0.15$.

Friction on A: $f_A = 0.15(40)(9.81) = 58.9$ N

Friction on B: $f_B = 0.15(40)(9.81)\cos 30° = 51.0$ N

Weight component of B down the slope: $40(9.81)\sin 30° = 196.2$ N

System: $196.2 - 51.0 - 58.9 = 80a$

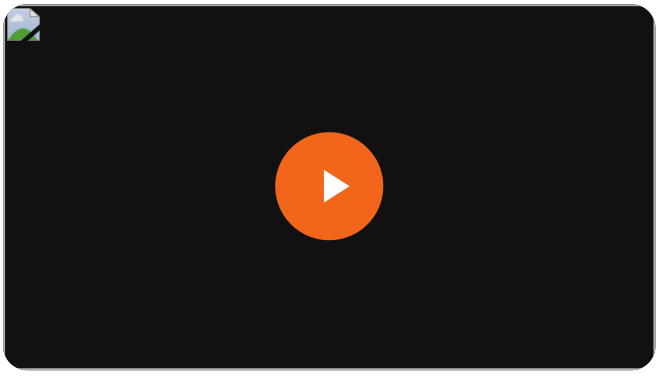
$a = 86.3 / 80 =$ **1.1 m s^{-2}**

For A: $T - f_A = m_A a \Rightarrow T = 58.9 + 40(1.08)$

$T = 102$ N = **0.10 kN**

Physics in Real Life

Real-world videos for this chapter. Tap a picture to play it here.



REAL LAUNCH FOOTAGE

NASA PACE Falcon 9 launch · NASA

Open in YouTube ↗

A rocket launch

The rocket pushes exhaust gas downwards and the gas pushes the rocket upwards (Newton's third law); thrust is the rate of change of momentum of the gas. A rocket under thrust is not a projectile.